The Data Solutions Company

Predicting Olist's Delivery Time

Trust Powered by Smart Technology

DSB - FT3

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Agenda



01 The business Problem

02 Data Analysis

03 model development

04 real-world customer journey

The Probelm

Olist’s delivery estimates are conservative, which may discourage customers from completing their purchases, potentially leading to lost sales.

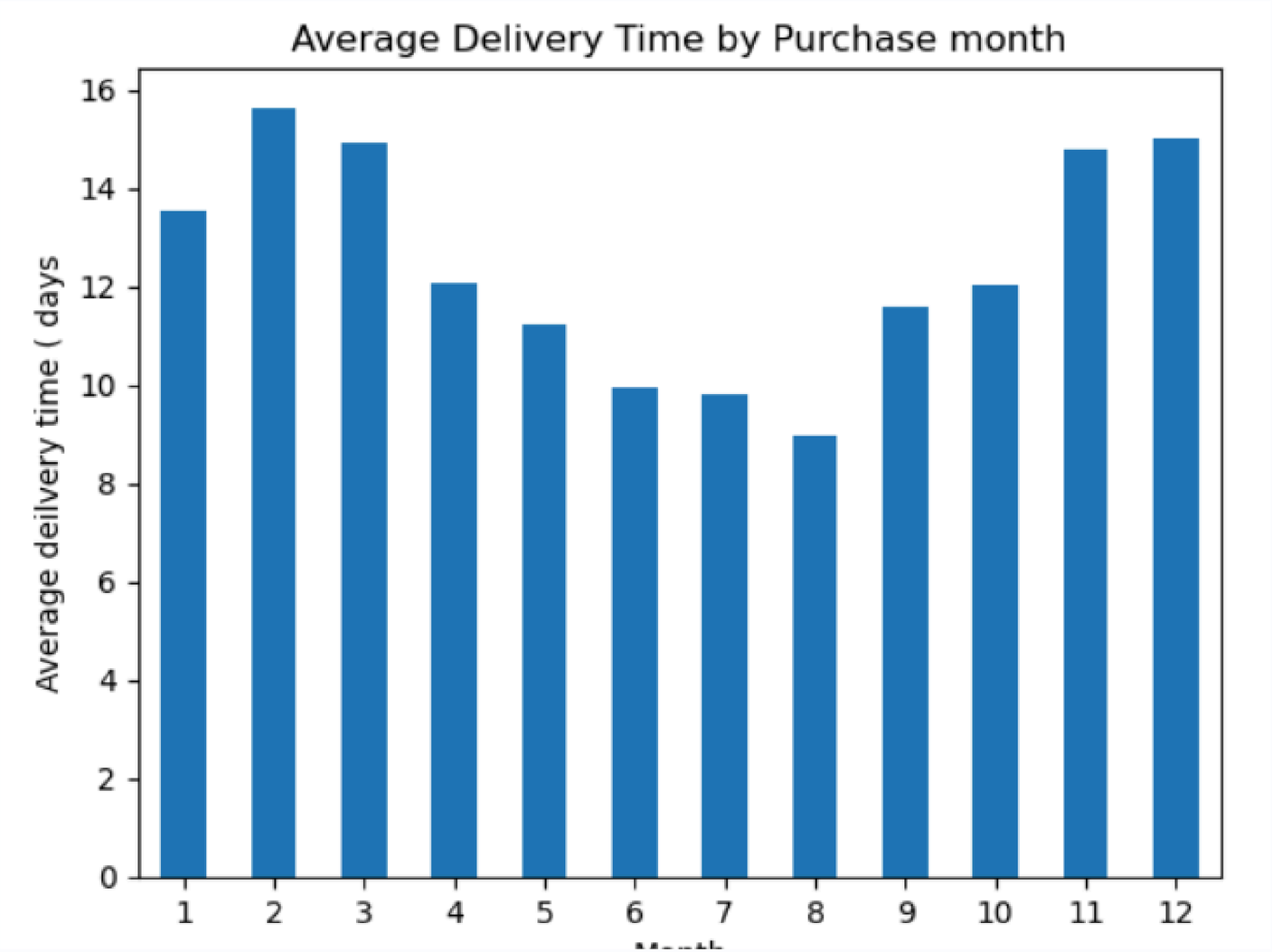
Our objective is to predict delivery time in days using historical order data, including product characteristics, order value, freight costs, and distance, to support more accurate delivery estimates and reduce potential cart abandonment.

How long do deliveries usually take?



Most Orders arrive fast, there is a small handful of unlucky orders that take a very long time.

Does the time of year matter?

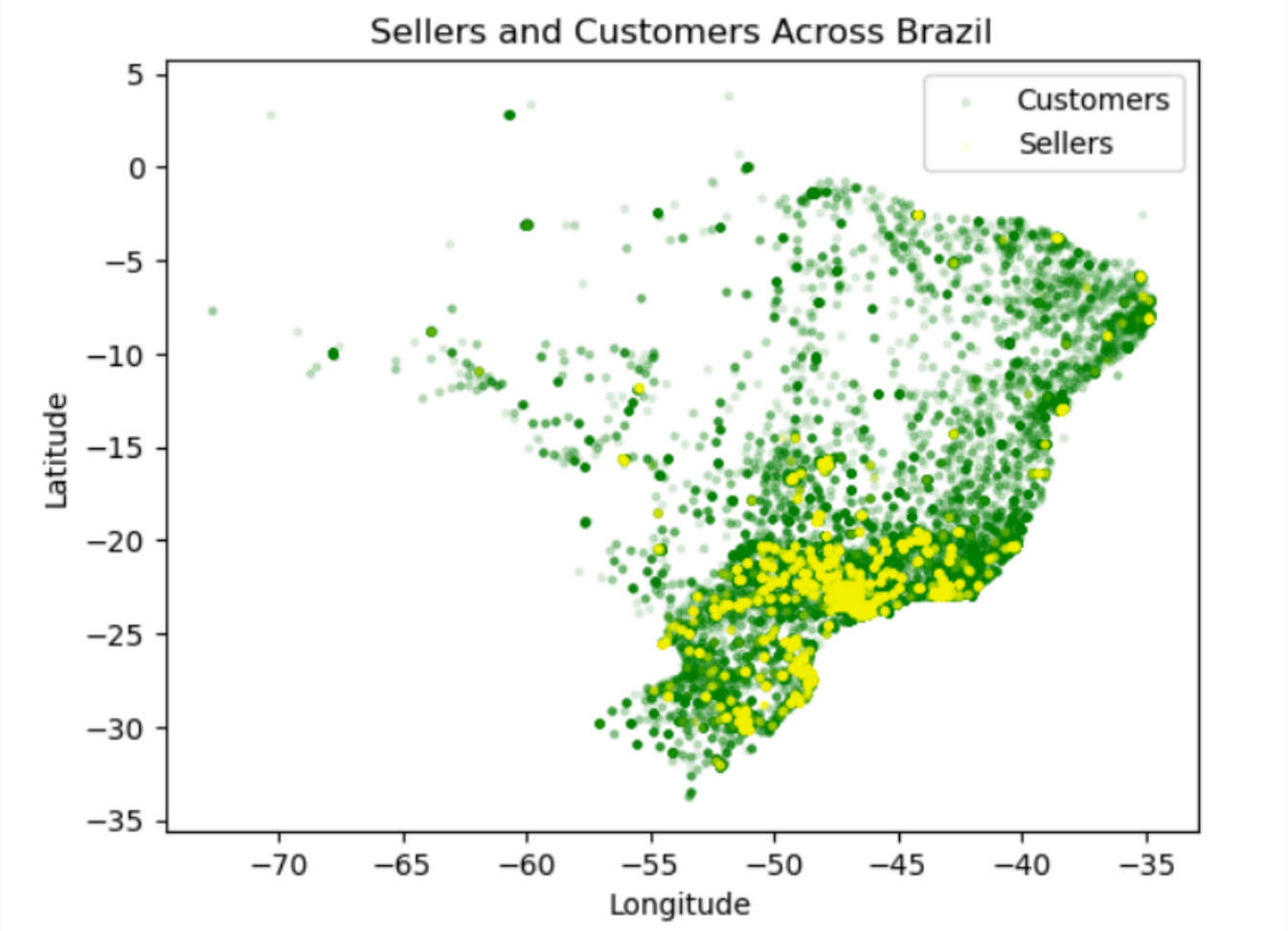


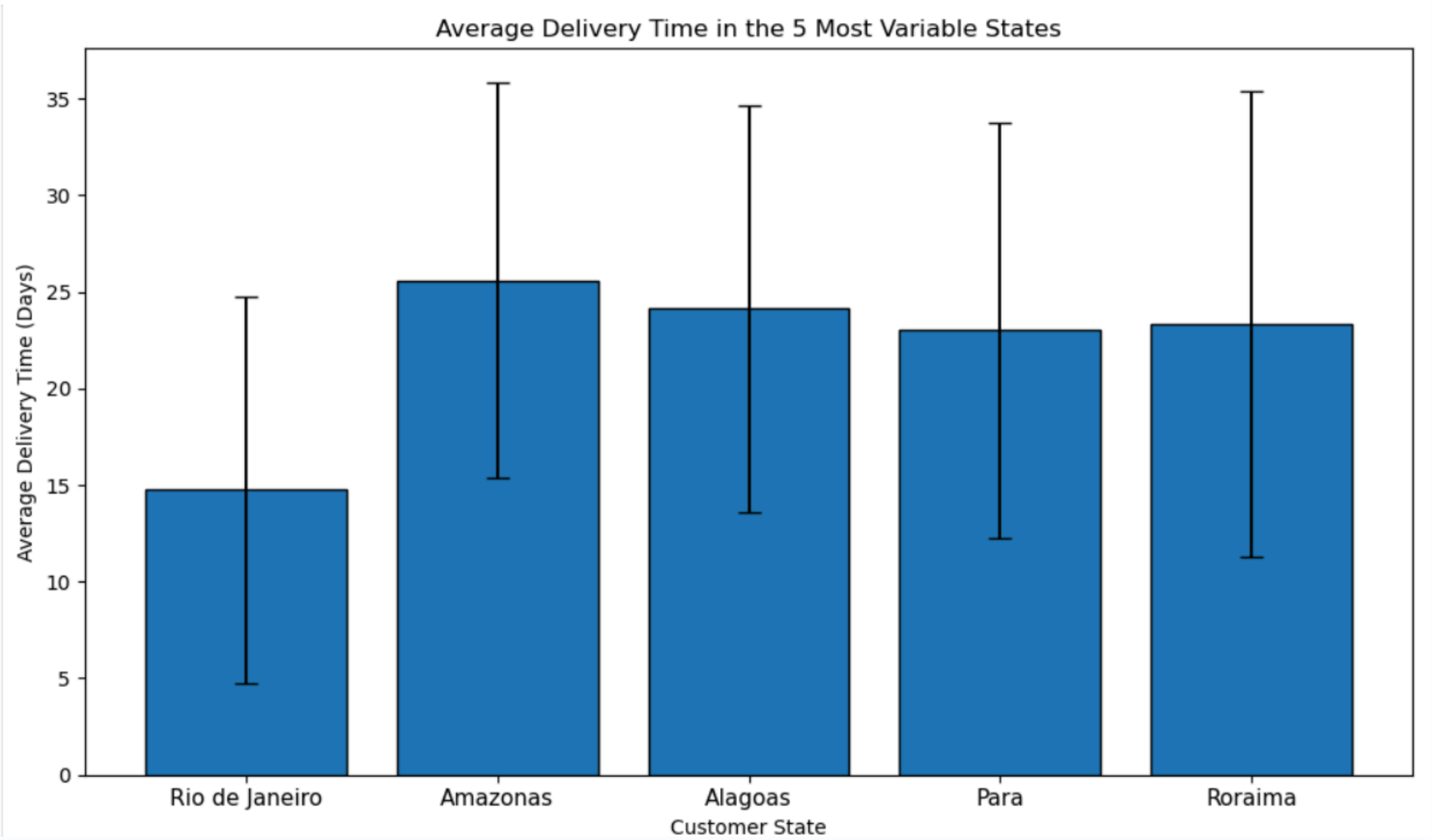
deliveries are slower in the first few months of the year and the last few months of the year, while the middle of the year (spring and summer) is faster.

Olist’s Geographic Network

Distance between sellers and buyers is highly correlated with the number of delivery dates.

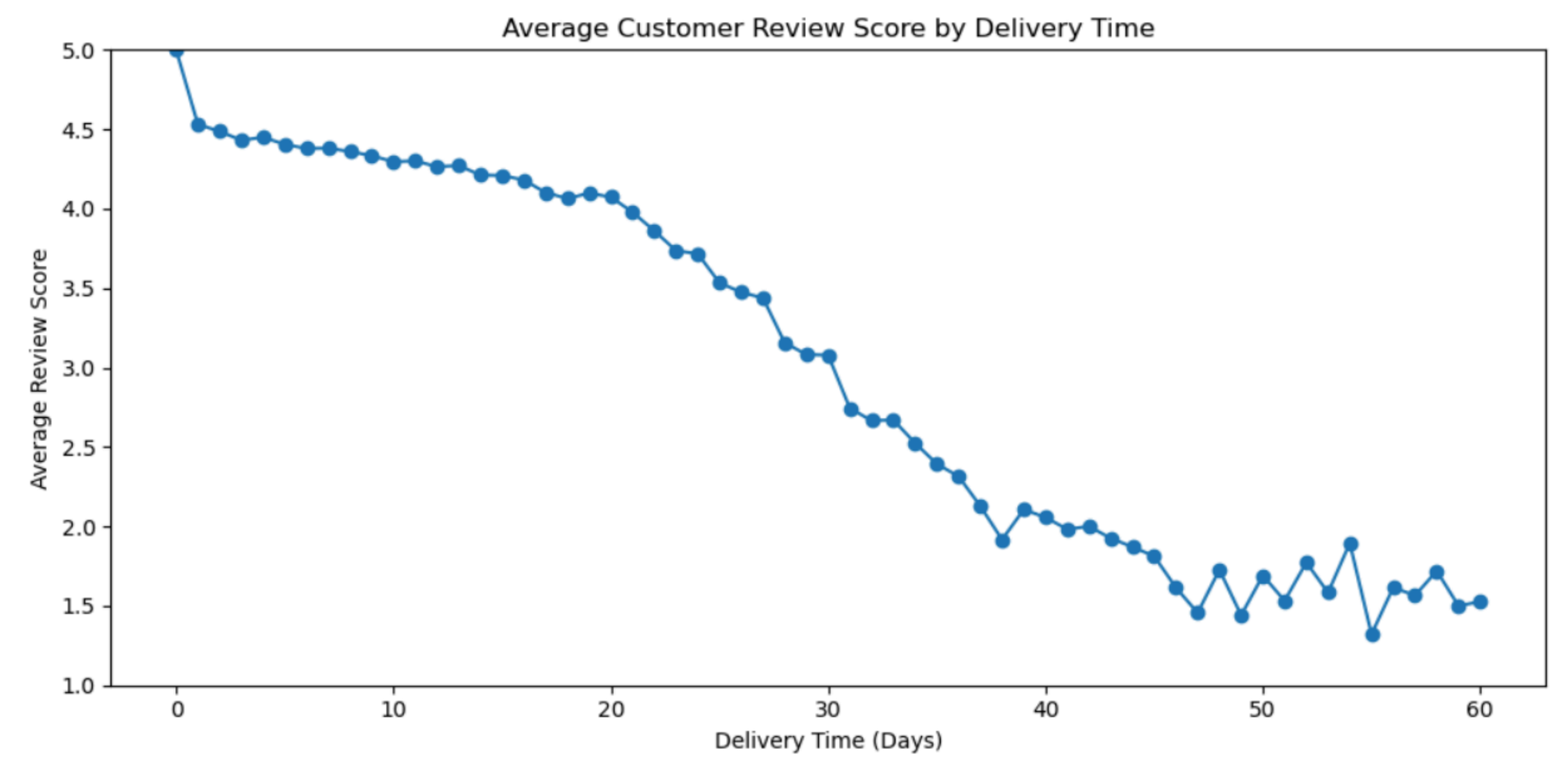
0.43





Delivery times are not consistent in each state

Customer Reviews with Delivery time



Data Cleaning & Preprocessing

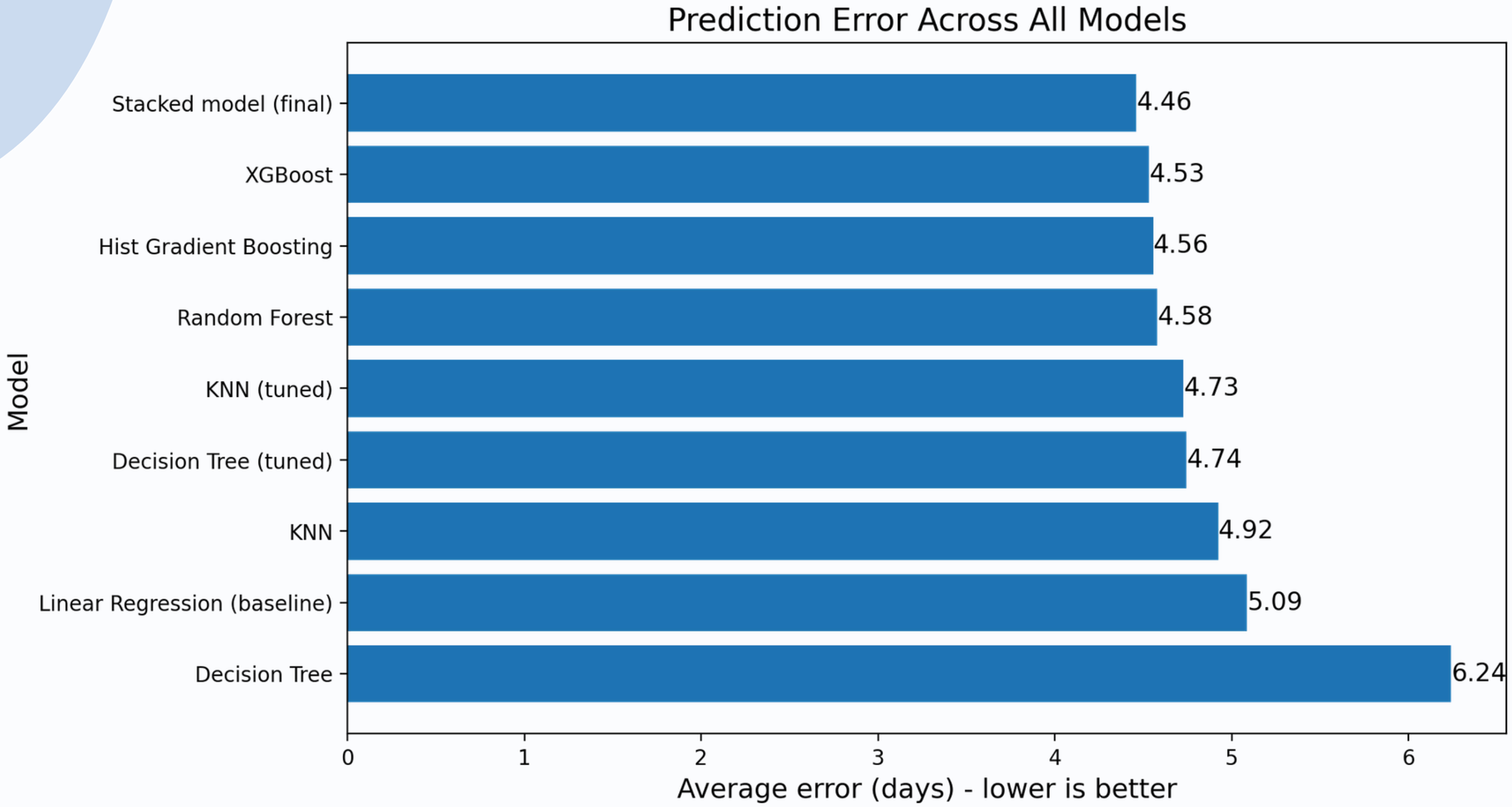
- - **Data Integration:** Combined order, customer, product, seller, location, reviews information into one using Olist's dataset.
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- - **Data Transformation:** Organised the data so each row represents one order and calculated delivery time in days, calculated new features such distance and days.
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- - **Missing Values:** Filled in missing information using imputation methods and removed data outside of brazil
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- - **Data Encoding & Scaling:** Converted categorical information into a format the models could understand and scaled numerical features to support model training.

How we measure our models

We chose MAE because it tells us exactly how many days off our predictions are, which can easily be interpreted by anyone

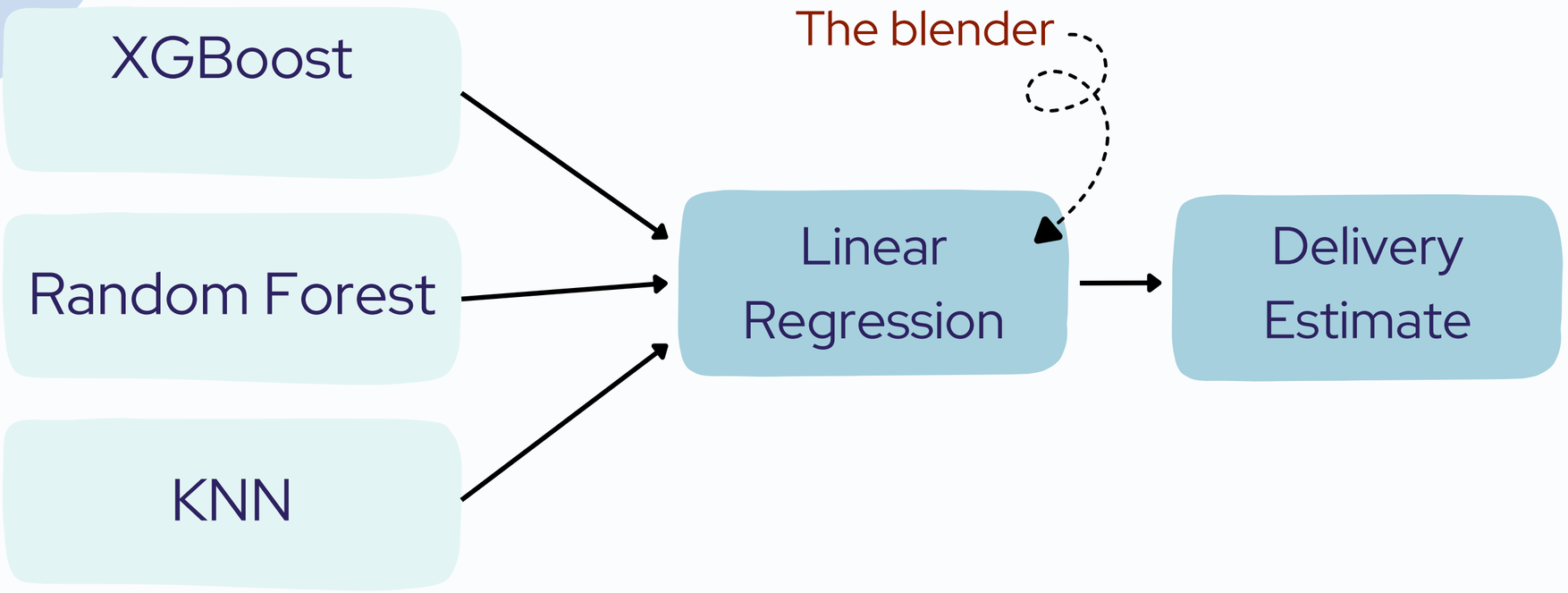
1. Calculate the errors for all predictions (predicted - actual)
2. Take the absolute value, so both the early and late count
3. Take the average

Model Comparison

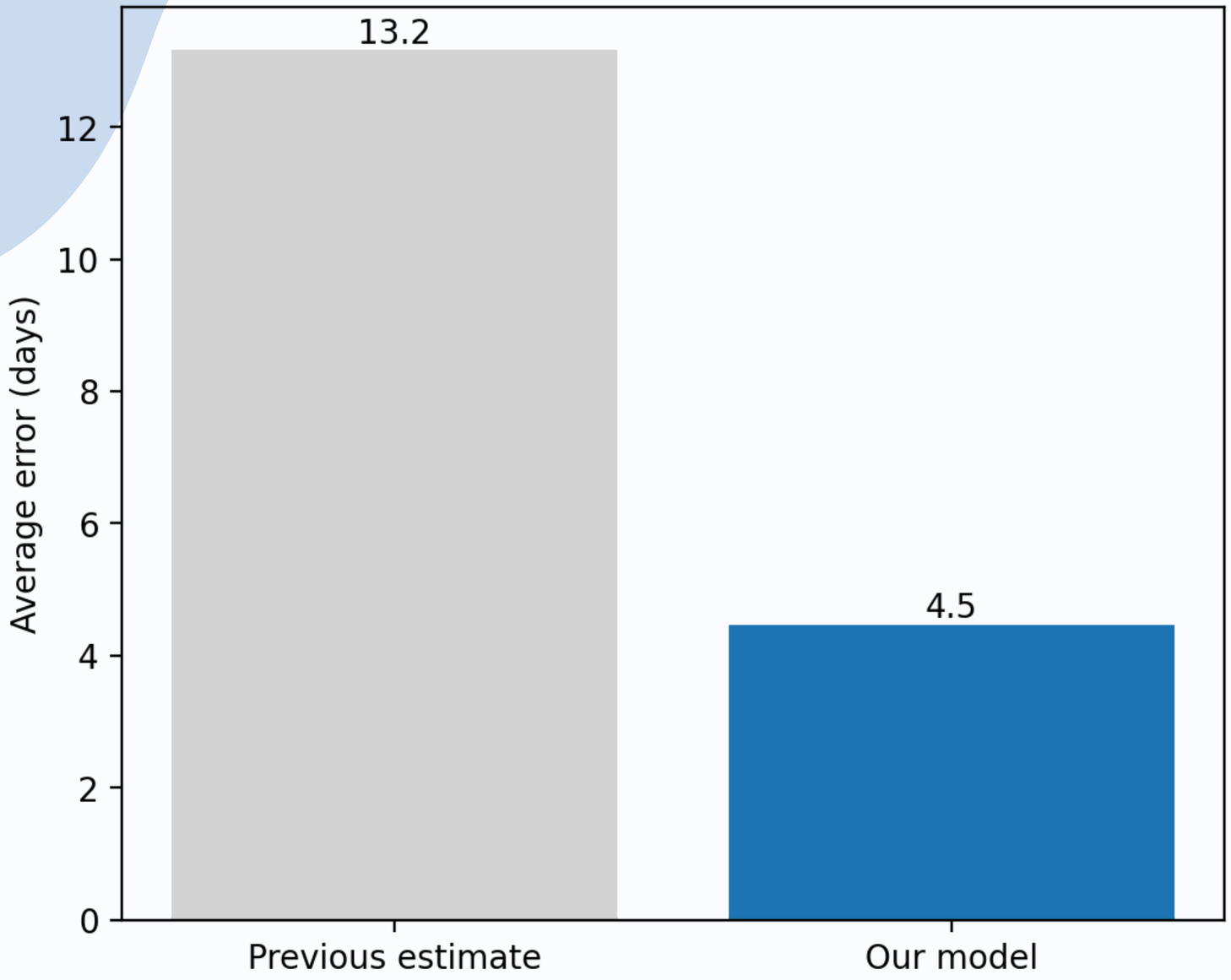


- We tested 8 Models and combined the best ones to get the lowest error : 4.5 days

Why Stacking was Chosen



- Three different models each make a prediction
- A simple blender combines them into one estimate
- Lowest error of all models tested: 4.5 days (simple baseline: 5 days)



our model is better by

66 %

Now lets see the model in action !